

RAGHUMANI MEHTA

Vancouver, BC

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Education

University of British Columbia

B.Sc. in Computer Science, Minor in Physics

- Dean's List (2024–2025)

Expected May 2028

Vancouver, BC

Technical Skills

Languages: Java, Python, C, C++, TypeScript, JavaScript, SQL, Rust

Frameworks & Tools: Git, Docker, GitHub Actions, Node.js, Express, Spring Boot, ROS, FreeRTOS, Pico SDK

Systems & Cloud: Linux/Unix, POSIX, Sockets, Networking, AWS (Lambda, S3, SQS, SNS, RDS, Textract)

Testing & Debugging: JUnit, Mocha/Chai, Pytest, GDB, Valgrind, OpenOCD, Raspberry Pi Debug Probe

AI-Assisted Development: GitHub Copilot, Codex, opencode, Copilot PR Reviews

Experience

Systopia Lab, UBC

May 2026 – Present

Undergraduate Research Assistant

Vancouver, BC

- Researching paging-system designs for protecting confidential cloud applications from memory access pattern leakage by a malicious hypervisor.
- Surveying confidential computing, user-level paging, and page-fault handling systems to evaluate implementation tradeoffs on Linux.
- Developing implementation specifications for a paging-based defense that can obscure memory access patterns while minimizing changes to guest applications.

UBC Sailbot

Feb. 2025 – Present

Software Lead

Vancouver, BC

- Lead a 30-member software team across 5 subteams, coordinating development, code reviews, integration planning, and technical direction.
- Design autonomy software architecture and align software interfaces with electrical and mechanical subsystem constraints.
- Use Dockerized Linux workflows, GitHub Actions, documentation, and code reviews to support reliable multi-team development.
- Built mock testing and simulation workflows to validate navigation logic before hardware integration.

UBC Department of Computer Science

Sept. 2025 – Apr. 2026

Undergraduate Teaching Assistant

Vancouver, BC

- CPSC 317: Computer Networking (Jan. 2026 – Apr. 2026): Led tutorials and supported students on IP forwarding, UDP, TCP, sockets, encryption, client-server programming, and debugging C/Java networking assignments.
- CPSC 310: Software Engineering (Sept. 2025 – Dec. 2025): Mentored 50+ students on TypeScript, Node.js, testing, CI workflows, and large software projects.

Projects

OCR System | *Python, AWS, PostgreSQL, REST APIs*

Jan. 2025 – Apr. 2025

- Led backend development for a serverless OCR workflow using AWS Textract, Lambda, S3, SQS, SNS, and RDS.
- Designed upload, OCR invocation, parsing, validation, user confirmation, and structured database storage workflows.
- Supported authenticated web and REST API access while coordinating backend quality across two development sprints.
- Tested the system with 50 simultaneous users and reviewed pull requests, merges, and backend integration changes.
- Repository: github.com/CPSC319-2025/Team-3-CCS

FreeRTOS Scheduler Extensions | *C, FreeRTOS, Real-Time Systems, Raspberry Pi Pico H*

Feb. 2026 – Apr. 2026

- Extended the FreeRTOS kernel with EDF, SRP, CBS, Global EDF, and Partitioned EDF scheduling policies.
- Implemented runtime stack sharing for SRP to reduce stack overhead while preserving resource-access correctness.
- Enabled and tested multicore scheduling behavior on Raspberry Pi Pico H using task timelines and resource-access traces.
- Used OpenOCD, GDB, and Raspberry Pi Debug Probe for low-level debugging and scheduler validation.
- Repository: github.com/alberto-escobar/CPSC538G